Math 142, S14 **Name:** Answer

Exam 2 **Section:** **SSN: xxx-xx-**

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	24	
3	20	
4	20	
5	18	
Total	100	

Good Luck !

Problem 1. (18 points) Determine whether the sequence is convergent or divergent. If it is convergent, find its limit.

(a) $a_n = \dfrac{n^2+n}{e^n}$

Let $f(x) = \dfrac{x^2+x}{e^x}$

$$\lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} \frac{x^2+x}{e^x} = \lim_{x\to+\infty} \frac{2x+1}{e^x}$$

$$= \lim_{x\to+\infty} \frac{2}{e^x} = 0$$

$$\Rightarrow \lim_{n\to+\infty} \frac{n^2+n}{e^n} = 0 \Rightarrow \{a_n\} \text{ Converges.}$$

(b) $a_n = \dfrac{2+n^3}{5-n^2+6n^3}$

$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} \frac{2+n^3}{5-n^2+6n^3} = \lim_{n\to+\infty} \frac{(2+n^3)/n^3}{(5-n^2+6n^3)/n^3}$$

$$= \lim_{n\to+\infty} \frac{\frac{2}{n^3}+1}{\frac{5}{n^3}-\frac{1}{n}+6}$$

$$= \frac{1}{6} \Rightarrow \{a_n\} \text{ Converges.}$$

(c) $a_n = (-1)^n \cos(\frac{1}{n})$

[Notice the change of sign of $(-1)^n \cos(\frac{1}{n})$]

$$\lim_{n\to+\infty} a_{2n} = \lim_{n\to+\infty} (-1)^{2n} \cos\left(\frac{1}{2n}\right) = \cos 0 = 1$$

$$\lim_{n\to+\infty} a_{2n+1} = \lim_{n\to+\infty} (-1)^{2n+1} \cos\left(\frac{1}{2n+1}\right) = -\cos 0 = -1$$

$$\Rightarrow \{a_n\} \text{ is divergent.}$$

Problem 2. (24 points) Determine whether the series is convergent or divergent.

(a) $\displaystyle\sum_{n=1}^{+\infty} \frac{2^n}{3^n}$ (If convergent, also find its sum for this one)

$\frac{2}{3} < 1 \Rightarrow \displaystyle\sum_{n=1}^{+\infty} \frac{2^n}{3^n}$ converges

and

$$\sum_{n=1}^{+\infty} \frac{2^n}{3^n} = \frac{\frac{2}{3}}{1-\frac{2}{3}} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2$$

$\hookrightarrow \displaystyle\sum_{n=1}^{+\infty} \frac{2}{3}\left(\frac{2}{3}\right)^{n-1}$

(b) $\displaystyle\sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^2}$

Let $f(x) = \frac{1}{x(\ln x)^2}$, clearly $f(x)$ decreases at $[2, +\infty)$.

$$\int_2^{+\infty} f(x)\,dx = \int_2^{+\infty} \frac{1}{x(\ln x)^2}\,dx = \lim_{t\to+\infty} \int_2^t \frac{1}{x(\ln x)^2}\,dx$$

$$\overset{u=\ln t}{=\!=} \lim_{t\to+\infty} \int_{\ln 2}^{\ln t} \frac{1}{u^2}\,du$$

$$= \lim_{t\to+\infty} \left(-\frac{1}{u}\right)\Big|_{\ln 2}^{\ln t} = \lim_{t\to+\infty}\left(-\frac{1}{\ln t} + \frac{1}{\ln 2}\right) = \frac{1}{\ln 2}$$

$\Rightarrow \displaystyle\sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^2}$ Converges.

(c) $\displaystyle\sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1}$ (use the limit Compasion Test)

Let $a_n = \frac{\sqrt{n^2+2}}{n^2+n+1}$, $b_n = \frac{1}{n}$.

$$\lim_{n\to+\infty} \frac{a_n}{b_n} = \lim_{n\to+\infty} \frac{\frac{\sqrt{n^2+2}}{n^2+n+1}}{\frac{1}{n}} = \lim_{n\to+\infty} \frac{\sqrt{n^2+2}\cdot n}{n^2+n+1}$$

$$= \lim_{n\to+\infty} \frac{(\sqrt{n^2+2})\cdot n/n^2}{(n^2+n+1)/n^2}$$

$$= \lim_{n\to+\infty} \frac{\sqrt{n^2+2}/n}{1+\frac{1}{n}+\frac{1}{n^2}} = \lim_{n\to+\infty} \frac{\sqrt{1+\frac{2}{n^2}}}{1+\frac{1}{n}+\frac{1}{n^2}} = \frac{\sqrt{1}}{1} = 1.$$

$\displaystyle\sum_{n=1}^{+\infty} b_n = \sum_{n=1}^{+\infty} \frac{1}{n}$ is divergent $\Rightarrow \displaystyle\sum_{n=1}^{+\infty} b_n = \sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1}$ is divergent.

Problem 3. (20 points) Determine whether the series is conditionally convergent, absolutely convergent or divergent.

(a) $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$

Consider $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$. Let $b_n = \frac{1}{\ln n}$

i) clearly $b_{n+1} = \frac{1}{\ln(n+1)} < b_n = \frac{1}{\ln n}$

ii) $\lim_{n\to+\infty} \frac{1}{\ln n} = 0$

Thus, by the Alternating Series Test, $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is convergent.

Now, Consider $\sum_{n=2}^{+\infty} \left|\frac{(-1)^n}{\ln n}\right| = \sum_{n=2}^{+\infty} \frac{1}{\ln n}$.

clearly $\frac{1}{\ln n} > \frac{1}{n}$ when $n \geq 2$

$\sum_{n=2}^{+\infty} \frac{1}{n}$ is divergent $\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{\ln n}$ is divergent by the Comparison Test.

Thus, $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.

(b) $\sum_{n=1}^{+\infty} \frac{(-1)^n e^{\frac{1}{n}}}{n^2}$

Let $a_n = \frac{e^{\frac{1}{n}}}{n^2}$, $b_n = \frac{1}{n^2}$

$\lim_{n\to+\infty} \frac{a_n}{b_n} = \lim_{n\to+\infty} \frac{\frac{e^{\frac{1}{n}}}{n^2}}{\frac{1}{n^2}} = \lim_{n\to+\infty} \frac{e^{\frac{1}{n}}}{n^2} \cdot n^2 = \lim_{n\to+\infty} e^{\frac{1}{n}} = e^0 = 1$

Since $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent, we get $\sum_{n=1}^{+\infty} \frac{e^{\frac{1}{n}}}{n^2}$ is also convergent by the Limit Comparison Test.

$\Rightarrow \sum_{n=1}^{+\infty} \left|\frac{(-1)^n e^{\frac{1}{n}}}{n^2}\right| = \sum_{n=1}^{+\infty} \frac{e^{1/n}}{n^2}$ is convergent.

$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^n e^{1/n}}{n^2}$ is absolutely convergent.

Problem 4. (20 points) Find the radius of convergence and interval of convergence of the series.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n2^n}$

Let $a_n = \frac{(-1)^n X^n}{n 2^n}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{(-1)^{n+1} X^{n+1}}{(n+1) 2^{n+1}}}{\frac{(-1)^n X^n}{n 2^n}}\right| = \left|\frac{X^{n+1}}{(n+1) 2^{n+1}} \cdot \frac{n 2^n}{X^n}\right| = \left|\frac{n}{n+1} \cdot \frac{X}{2}\right|$$

$$= \left|\frac{1}{1+1/n} \cdot \frac{X}{2}\right| \to \left|\frac{X}{2}\right| \text{ as } n \to +\infty$$

$$\Rightarrow \left|\frac{X}{2}\right| < 1 \Rightarrow |X| < 2 \Rightarrow R = 2.$$

For $X=2$, we have $\sum_{n=1}^{+\infty} \frac{(-1)^n 2^n}{n \cdot 2^n} = \sum_{n=1}^{+\infty} \frac{(-1)^n}{n}$ converges

$X=-2$, we have $\sum_{n=1}^{+\infty} \frac{(-1)^n (-2)^n}{n \cdot 2^n} = \sum_{n=1}^{+\infty} \frac{2^n}{n \cdot 2^n} = \sum_{n=1}^{+\infty} \frac{1}{n}$ divergens

Thus the interval of convergence is $(-2, 2]$.

(b) $\sum_{n=0}^{+\infty} \sqrt{n}(x-1)^n$

let $a_n = \sqrt{n}(X-1)^n$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\sqrt{n+1}(X-1)^{n+1}}{\sqrt{n}(X-1)^n}\right| = \left|\sqrt{\frac{n+1}{n}} \cdot (X-1)\right| = \left|\sqrt{1+\frac{1}{n}}(X-1)\right|$$

$$\to |X-1| \text{ as } n \to +\infty$$

$$\Rightarrow |X-1| < 1 \Rightarrow R = 1.$$

$|X-1| < 1 \Rightarrow 0 < X < 2.$

For $X=0$, we have $\sum_{n=0}^{+\infty} \sqrt{n}(-1)^n$ diverges.

$X=2$, we have $\sum_{n=0}^{+\infty} \sqrt{n}(1)^n = \sum_{n=0}^{+\infty} \sqrt{n}$ diverges.

Thus the interval of convergence is $(0, 2)$.

Problem 5. (18 Points) Find the Maclaurin series (power series centered at 0) and its radius of convergence for the following functions.

(a) $f(x) = \dfrac{1}{2}(e^x + e^{-x})$

$$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!} \qquad e^{-x} = \sum_{n=0}^{+\infty} \frac{(-x)^n}{n!} = \sum_{n=0}^{+\infty} \frac{(-1)^n x^n}{n!}$$

$$f(x) = \frac{1}{2}(e^x + e^{-x}) = \frac{1}{2}\left(\sum_{n=0}^{+\infty} \frac{x^n}{n!} + \sum_{n=0}^{+\infty} (-1)^n \frac{x^n}{n!}\right)$$

$$= \sum_{n=0}^{+\infty} \frac{1}{2}(1+(-1)^n)\frac{x^n}{n!} \qquad \frac{1}{2}(1+(-1)^n) = \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

$$= \sum_{n=0}^{+\infty} \frac{x^{2n}}{(2n)!}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{x^{2(n+1)}}{(2n+2)!} \cdot \frac{(2n)!}{x^{2n}}\right| = \left|\frac{x^2}{(2n+1)(2n+2)}\right| \to 0 \quad \text{as } n \to +\infty$$

$$\Rightarrow R = +\infty$$

(b) $f(x) = \dfrac{x^2}{1-2x}$

$$\frac{1}{1-2x} = \sum_{n=0}^{+\infty} (2x)^n = \sum_{n=0}^{+\infty} 2^n x^n$$

$$\Rightarrow f(x) = x^2 \cdot \sum_{n=0}^{+\infty} 2^n x^n$$

$$= \sum_{n=0}^{+\infty} 2^n x^{n+2}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{2^{n+1} x^{n+3}}{2^n x^{n+2}}\right| = |2x|$$

$$|2x| < 1 \Rightarrow |x| < \frac{1}{2} \Rightarrow R = \frac{1}{2}$$

Extra Credit. (6 points) Find the Maclaurin series for the function

$$f(x) = \cos^2\left(\frac{x}{2}\right).$$

(*Hint: Apply the half-angle identity first.*)

Notice $\cos^2\left(\frac{x}{2}\right) = \frac{\cos x + 1}{2}$

$$\cos x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!}.$$

$$\Rightarrow \cos^2\left(\frac{x}{2}\right) = \frac{1}{2}\left[\left(\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!}\right) + 1\right]$$

$$= \left(\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{2(2n)!}\right) + \frac{1}{2}$$

$$= 1 + \sum_{n=1}^{+\infty} (-1)^n \frac{x^{2n}}{2(2n)!}$$

$$= 1 - \frac{x^2}{2 \cdot 2!} + \frac{x^4}{2 \cdot 4!} - \frac{x^6}{2 \cdot 6!} + \cdots$$